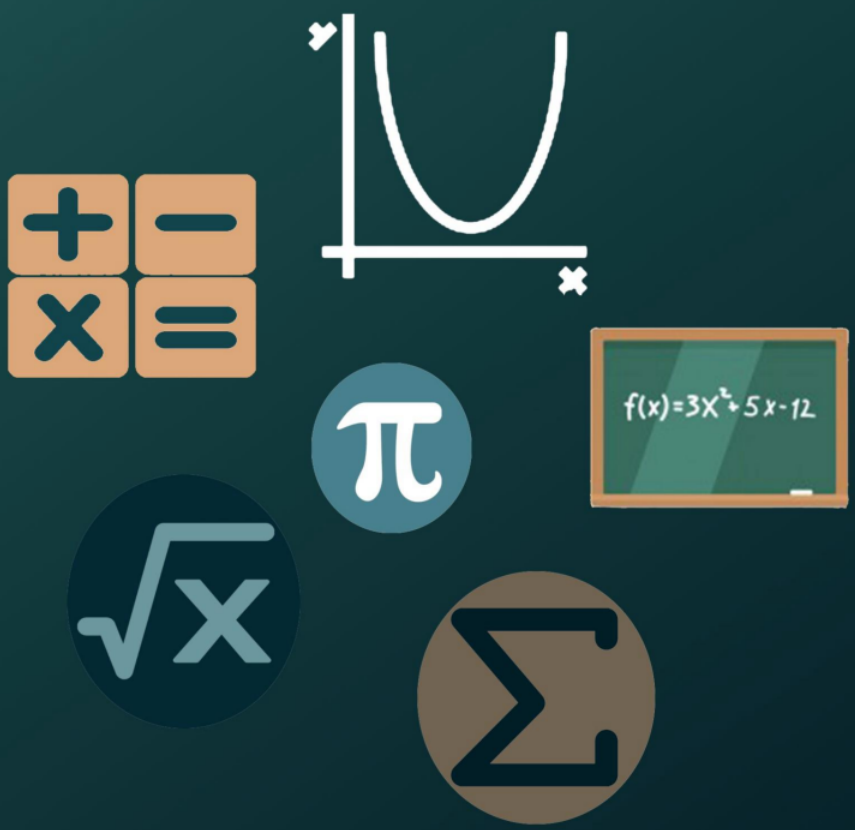
MATHEMATICS

Differentiability



SHEET

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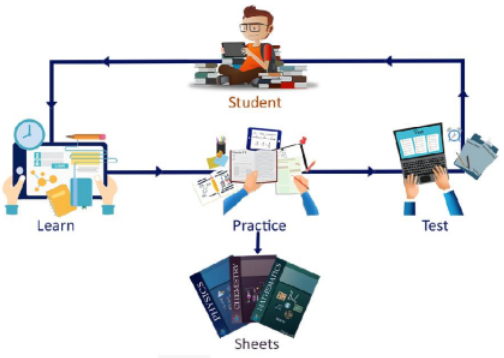
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Concept of Derivative				
2	Existence of Derivative	1 - 6	12, 13, 16 - 18	1 - 2	26
3	Derivability & Continuity	7 - 10	1 - 3, 6 - 8, 9 - 11, 14, 19, 20	3 - 8	32
4	Algebra of Differentiability	11 - 12	4, 5, 21 - 24	9 - 14	4 - 8, 14 -16, 17, 20 - 22
5	Important Concepts	13	15	15	1 - 3, 11 - 13, 19, 27 - 29
6	Determination of Function satisfying the Functional Rule			16	9 - 10, 18, 30
7	After complete chapter				23 - 26, 31

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Let $f(x) = \begin{cases} (x-1)\sin\dfrac{1}{x-1}, & \text{if } x \neq 1 \\ 0, & \text{if } x = 1 \end{cases}$

then which one of the following is true.

(A) f is differentiable at x = 0 and at x = 1.

(B) f is differentiable at x = 0 but not at x = 1.

(C) f is differentiable at x = 1 but not at x = 0.

(D) f is neither differentiable at x = 0 nor at x = 1.

Q2. If $f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then f(x) is -

(A) Continuous as well as differentiable for all x

(B) Continuous for all x but not differentiable at x = 0

(C) Neither differentiable nor continuous at x = 0

(D) Discontinuous every where

Q3. The function $f(x) = |x| + |x-1|$ is -

(A) Continuous at x = 1, but not differentiable at x = 1

(B) Both continuous and differentiable at x = 1

(C) Not continuous at x = 1

(D) Not differentiable at x = 1

Q4. Let $f: R \to R$ be a function defined by f(x) = Min $\{x + 1, |x| + 1\}$. Then which of the following is true -

(A) $f(x) \geq 1$ for all $x \in R$

(B) f(x) is not differentiable at x = 1

(C) f(x) is differentiable everywhere

(D) f(x) is not differentiable at x = 0

Q5. Let $f(x) = \dfrac{1}{1+e^{\frac{1}{x}}}$ for $x \neq 0$ and $f(0) = 0$, then

(A) $f(0^+)$ does not exist

(B) $f(0^-)$ is equal to zero

(C) $f'(0^-)$ is equal to 1

(D) $f'(0^+)$ is equal to zero

Q6. Which of the following functions defined below are NOT differentiable at the indicated point?

(A) $f(x) = \begin{cases} x^2 & \text{if } -1 \leq x < 0 \\ -x^2 & \text{if } 0 \leq x \leq 1 \end{cases}$ at $x = 0$

(B) $g(x) = \begin{cases} x & \text{if } -1 \leq x < 0 \\ \tan x & \text{if } 0 \leq x \leq \dfrac{\pi}{4} \end{cases}$ at $x = 0$

(C) $h(x) = \begin{cases} \sin 2x & \text{if } x \leq 0 \\ 2x & \text{if } x > 0 \end{cases}$ at $x = 0$

(D) $k(x) = \begin{cases} x & \text{if } 0 \leq x \leq 1 \\ 2-x & \text{if } 1 < x \leq 2 \end{cases}$ at $x = 1$

Q7. For $x > 0$, let $h(x) = \begin{cases} \frac{1}{q} & \text{if } x = \frac{p}{q} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$

where p & q > 0 are relatively prime integers then which one does not hold good?

(A) h(x) is discontinuous for all x in $(0, \infty)$

(B) h(x) is continuous for each irrational in $(0, \infty)$

(C) h(x) is discontinuous for each rational in $(0, \infty)$

(D) h(x) is not derivable for all x in $(0, \infty)$.

Q8. If $f(x) = \sin^{-1}(\sin x)$; $x \in R$ then f is

(A) continuous and differentiable for all x

(B) continuous for all x but not differentiable for all $x = (2k+1)\frac{\pi}{2}$, $k \in I$

(C) neither continuous nor differentiable for $x = (2k-1)\frac{\pi}{2}$; $k \in I$

(D) neither continuous nor differentiable for $x \in R - [-1, 1]$

Q9. If $f(x) = a|\sin x| + be^{|x|} + c|x|^3$ and if $f(x)$ is differentiable at $x = 0$ then

(A) $b = 0, c = 0$, a is any real

(B) $a = 0, b = 0$, c is any real

(C) $c = 0, a = 0$, b is any real

(D) $a + b = 0$

Q10. The number of values of a for the which the function $f(x) = (x+1)|x-a|$ is differentiable $\forall x \in R$, is

(A) 0 (B) 1

(C) 2 (D) more than 2

Q11. Which one of the following is not true always?

(A) if f(x) is not continuous at x = a, then it is not differentiable at x = a.

(B) If f(x) is continuous at x = a, then it is differentiable at x = a.

(C) if f(x) and g(x) are differentiable at x = a. then f(x) + g(x) is also differentiable at x = a.

(D) if a function f(x) is continuous at x = a, then $\lim_{x \to a} f(x)$ exists.

Q12. Let f be a real valued function defined on R given by $f(x) = \{x\} + 2[x]$.

Where [y] and { y } denote largest integer $\leq$ y and fractional part of y respectively.

The function f(x) is

(A) continuous and differentiable $\forall x \in R$.

(B) continuous $\forall x \in R$ but not differentiable at integral points.

(C) neither continuous nor derivable at integral points.

(D) nowhere differentiable.

Q13. The function $f(x) = \begin{cases} 2x+1 & , x \in Q \\ x^2 - 2x + 5 & , x \notin Q \end{cases}$ is

(A) continuous no where

(B) differentiable no where

(C) continuous but not differentiable exactly at one point

(D) differentiable and continuous only at one point and discontinuous elsewhere

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. $f(x) = |x[x]|$ in $-1 \leq x \leq 2$, where [x] is greatest integer $\leq$ x then f(x) is :-

(A) continuous at x = 0

(B) discontinuous at x = 0

(C) not differentiable at x = 2

(D) differentiable at x = 2

Q2. If $f(x) = 2 + |\sin^{-1} x|$, it is :

(A) continuous no where

(B) continuous everywhere in its domain

(C) differentiable no where in its domain

(D) Not differentiable at $x = 0$

Q3. Let $f(x) = \begin{cases} x^2 \cos\frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin\frac{1}{x}, & x > 0 \end{cases}$, then which of the following is (are) correct ?

(A) f(x) is continuous but not differentiable at $x = 0$

(B) f(x) is continuous and differentiable at $x = 0$

(C) f'(x) is continuous but not differentiable at $x = 0$

(D) f'(x) is discontinuous at $x = 0$

Q4. $f(x) = 1 + x.[\cos x]$ in $0 < x \le \pi/2$, where [ ] denotes greatest integer function then :-

(A) It is continuous in $0 < x < \pi/2$

(B) It is differentiable in $0 < x < \pi/2$

(C) Its maximum value is 2

(D) It is not differentiable in $0 < x < \pi/2$

Q5. $f(x) = |x| + |\sin x|$ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. It is :-

(A) Continuous no where

(B) Continuous every where

(C) Differentiable no where

(D) Differentiable everywhere except at $x = 0$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 6 to 8)

Consider the function $f : R \to R$, defined as

$$f(x) = \begin{cases} x^2 - x + 3, & \text{if } x \in Q \text{ in } (-\infty, 3) \\ x + a, & \text{if } x \notin Q \text{ in } (-\infty, 2) \\ 2^x + 1, & \text{if } x \notin Q \text{ in } (2, 3) \\ 9\tan\left(\frac{\pi x}{12}\right), & \text{if } x \ge 3 \end{cases}$$

Q6. If f(x) is continuous at $x = 2$ then the value of a is

(A) 1 (B) 2

(C) 3 (D) indeterminate

Q7. The function $f(x)$ at $x = 3$

(A) has non-removable discontinuity.

(B) has removable discontinuity.

(C) is differentiable.

(D) is continuous but not differentiable

Q8. $f'(4)$ is equal to

(A) π (B) 3π (C) $\frac{3\pi}{2}$ (D) $\frac{3\pi}{16}$

Paragraph # 2 (Ques. 9 to 11)

Consider two function $y = f(x)$ and $y = g(x)$ defined as

$$f(x) = \begin{cases} ax^2 + b, & 0 \le x \le 1 \\ 2bx + 2b, & 1 < x \le 3 \\ (a-1)x + 2a - 3, & 3 < x \le 4 \end{cases}$$

and $$g(x) = \begin{cases} cx^2 + d, & 0 \le x \le 2 \\ dx + 3 - c, & 2 < x < 3 \\ x^2 + b + 1, & 3 \le x \le 4 \end{cases}.$$

Q9. f(x) is continuous at x = 1 but not differentiable at x = 1, if

(A) a = 1, b = 0

(B) a = 1, b = 2

(C) a = 3, b = 1

(D) a and b are any integers

Q10. g(x) is continuous at x = 2, if

(A) c = 1, d = 2 (B) c = 2, d = 3

(C) c = 1, d = –1 (D) c = 1, d = 4

Q11. If f is continuous and differentiable at x = 3, then

(A) $a = -\frac{1}{3},\ b = \frac{2}{3}$

(B) $a = \frac{2}{3},\ b = -\frac{1}{3}$

(C) $a = \frac{1}{3},\ b = -\frac{2}{3}$

(D) $a = 2,\ b = \frac{1}{2}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q12.

Column - I	**Column - II**
(A) If f(x) is derivable at x = 3 & f'(3) = 2, then $\lim_{h\to 0}\frac{f(3+h^2)-f(3-h^2)}{2h^2}$ equals	(P) 0
(B) Let f(x) be a function satisfying the condition f(–x) = f(x) for all real x. If f'(0) exists, then its value is equal to	(Q) 1
(C) For the function $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ the derivative from the left f'(0⁻) equals	(R) 2
(D) The number of points at which the function f(x) = max. {a – x, a + x, b}, –∞ < x < ∞, 0 < a < b cannot be differentiable is	(S) 3

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q13. If $f(x) = \begin{cases} ax^2 - b & \text{if } |x| < 1 \\ -\frac{1}{|x|} & \text{if } |x| \geq 1 \end{cases}$ is derivable at x = 1. Find the value of a + b.

Q14. Let $g(x) = \begin{cases} a\sqrt{x+2}, & 0 < x < 2 \\ bx + 2, & 2 \leq x < 5 \end{cases}$. If g (x) is derivable on (0, 5), then find (2a + b).

Q15. Let $f(x) = [3 + 4 \sin x]$ (where [ ] denotes the greatest integer function). If sum of all the values of 'x' in $[\pi, 2\pi]$ where $f(x)$ fails to be differentiable, is $\frac{k\pi}{2}$, then find the value of k.

SUBJECTIVE TYPE

Q16. Find the left and right hand derivative of the following function at given point
(a) $f(x) = |\ln x|$ at $x = 1$
(b) $f(x) = e^{-|x|}$ at $x = 0$

Q17. If $f(x) = \begin{cases} ax + b, & x \le -1 \\ ax^3 + x + 2b, & x > -1 \end{cases}$ is differentiable for all $x \in R$ find 'a' & 'b'.

Q18. If $f(x) = \begin{cases} x^m . \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ is continuous but not differentiable at $x = 0$ then find m.

Q19. $f(x) = \begin{cases} |1 - 4x^2|, & 0 \le x < 1 \\ [x^2 - 2x], & 1 \le x \le 2 \end{cases}$, check differentiability in (0, 2), where [ ] denotes greatest integer function.

Q20. A function f is defined as follows :

$$f(x) = \begin{cases} 1 & \text{for } -\infty < x < 0 \\ 1 + |\sin x| & \text{for } 0 \le x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2 & \text{for } \frac{\pi}{2} \le x < +\infty \end{cases}$$

discuss continuity and differentiability at $x = 0$ and $x = \pi/2$.

Q21. Let $f(x) = \text{sgn } x$ and $g(x) = x(1 - x^2)$. Investigate the composite functions $f(g(x))$ and $g(f(x))$ for continuity and differentiability.

Q22. Discuss the differentiability of $f(x) =$ maximum $\{2 \sin x, 1 - \cos x\}$ $\forall$ $x \in (0, \pi)$.

Q23. Discuss the continuity & differentiability of the function $f(x) = \sin x + \sin|x|$, $x \in R$. Draw a rough sketch of the graph of f(x).

Q24. Examine the continuity and differentiability of $f(x) = |x| + |x - 1| + |x - 2|$ $x \in R$. Also draw the graph of f(x).

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Number of points of non-differentiability of the function
$g(x) = [x^2]\{\cos^2 4x\} + \{x^2\}[\cos^2 4x] + x^2 \sin^2 4x + [x^2][\cos^2 4x] + \{x^2\}\{\cos^2 4x\}$ in $(-50, 50)$
where [x] and {x} denotes the greatest integer function and fractional part function of x respectively, is equal to :-
(A) 98 (B) 99 (C) 100 (D) 0

Q2. If $f(x) = \text{sgn}(x) - \dfrac{2^{1/x}+1}{2^{1/x}-1}$, $x \neq 0$ (where sgn(x) denotes signum function) & $f(0) = 0$, then observe the following statements -

(I) $f(x)$ is continuous at $x = 0$

(II) $f(x)$ is discontinuous at $x = 0$

(III) RHD exists at $x = 0$

(IV) LHD exists at $x = 0$

The correct sequence of true/false of the foresaid statements is -

(A) TFTT (B) TFTF

(C) FTTT (D) FTTF

Q3. For what triplets of real numbers (a, b, c) with $a \neq 0$ the function

$$f(x) = \begin{bmatrix} x & x \leq 1 \\ ax^2 + bx + c & \text{otherwise} \end{bmatrix}$$

is differentiable for all real x ?

(A) $\{(a, 1-2a, a) \mid a \in R, a \neq 0\}$

(B) $\{(a, 1-2a, c) \mid a, c \in R, a \neq 0\}$

(C) $\{(a, b, c) \mid a, b, c \in R, a + b + c = 1\}$

(D) $\{(a, 1-2a, 0) \mid a \in R, a \neq 0\}$

Q4. Given

$$f(x) = \begin{bmatrix} \log_a(a^{[x]+[-x]})^x & \left(\dfrac{2a^{\left(\frac{\{x\}+\{-x\}}{|x|}\right)}-5}{3+a^{\frac{1}{|x|}}}\right) & \text{for } |x| \neq 0; a > 1 \\ 0 & & \text{for } x = 0 \end{bmatrix},$$

where [ ] represents the integral part function, then :

(A) f is continuous but not differentiable at $x = 0$

(B) f is continuous & differentiable at $x = 0$

(C) the differentiability of 'f' at $x = 0$ depends on the value of a

(D) f is continuous & differentiable at $x = 0$ and for $a = e$ only

Q5. If $f(x) = \begin{bmatrix} x + \{x\} + x\sin\{x\} & \text{for} & x \neq 0 \\ 0 & \text{for} & x = 0 \end{bmatrix}$

where $\{x\}$ denotes the fractional part function, then :-

(A) 'f' is continuous & differentiable at $x = 0$

(B) 'f' is continuous but not differentiable at $x = 0$

(C) 'f' is continuous & differentiable at $x = 2$

(D) None of these

Q6. If $2x + 3|y| = 4y$, then y as a function of x i.e. $y = f(x)$, is -

(A) discontinuous at one point

(B) non differentiable at one point

(C) discontinuous & non differentiable at same point

(D) continuous & differentiable everywhere

Q7. Consider

$$f(x) = \left[\frac{2(\sin x - \sin^3 x) + |\sin x - \sin^3 x|}{2(\sin x - \sin^3 x) - |\sin x - \sin^3 x|}\right]$$

$x \in (0, \pi) - \{\pi/2\}$, $f(\pi/2) = 3$ where [ ] denotes the greatest integer function then-

(A) f is continuous and differentiable at $x = \pi/2$

(B) f is continuous but not differentiable at $x = \pi/2$

(C) f is neither continuous nor differentiable at $x = \pi/2$

(D) None of these

Q8. Let $f(x) = x - x^2$ and

$$g(x) = \begin{cases} \max\{f(t), 0 \leq t \leq x\}; & 0 \leq x \leq 1 \\ \sin \pi x, & x > 1 \end{cases}.$$

Then in the interval $[0, \infty)$ -

(A) g(x) is everywhere continuous except at two points

(B) g(x) is everywhere differentiable except at two points

(C) g (x) is everywhere differentiable except at $x = 1$

(D) none of these

Q9. The set of all points where the function $f(x) = \frac{x}{1+|x|}$ is differentiable is :-

(A) $(-\infty, \infty)$

(B) $[0, \infty)$

(C) $(-\infty, 0) \cup (0, \infty)$

(D) none

Q10. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in I$ and p is a prime number. The number of points where f(x) is not differentiable is :-

(A) $p - 1$ (B) $p + 1$

(C) $2p + 1$ (D) $2p - 1$

Here [x] denotes greatest integer function.

Q11. The function $f(x) = (x^2 - 1) | x^2 - 3x + 2 | + \cos(|x|)$ is NOT differentiable at :

(A) –1 (B) 0

(C) 1 (D) 2

Q12. Consider the function

$f(x) = x^2 - |x^2 - 1| + 2 || x | - 1| + 2 | x | - 7$

Statement-1 :

f is not differentiable at x = 1, –1 and 0.

because

Statement-2 :

| x | is not differentiable at x = 0 and $| x^2 - 1|$ is not differentiable at x = 1 and –1.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation of statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q13. If $f(x) = [\sin x] + \sqrt{\sin x - [\sin x]}$, where [.] denotes greatest integer function, then $f(x)$ is -

(A) differentiable at $x = \frac{\pi}{2}$ & $x = \pi$ both

(B) differentiable at $x = \frac{\pi}{2}$ but not at $x = \pi$

(C) differentiable at $x = \pi$ but not at $x = \frac{\pi}{2}$

(D) neither differentiable at $x = \pi$ nor at $x = \frac{\pi}{2}$

Q14. Let $f(x) = \left[\frac{x + | x - 2 |}{2}\right]$ (where [.] denotes greatest integer function), then -

(A) $f(x)$ is discontinuous at every integer

(B) $f(x)$ is non-differentiable at every integer

(C) $f(x)$ is non differentiable at only one positive integer

(D) $f(x)$ is continuous $\forall\ x \in (-\infty, 3)$

Q15. Consider the functions $f(x) = x^2 - 2x$ and $g(x) = - | x |$

Statement-1 : The composite function $F(x) = f(g(x))$ is not derivable at x = 0.

because

Statement-2 : $F'(0^+) = 2$ and $F'(0^-) = -2$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation of statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q16. Let f be an injective and differentiable function such that $f(x) \cdot f(y) + 2 = f(x) + f(y) + f(xy)$ for all non negative real x and y with $f'(0) = 0$, $f'(1) = 2 \neq f(0)$, then :-

(A) $x f'(x) - 2f(x) + 2 = 0$

(B) $x f'(x) + 2f(x) - 2 = 0$

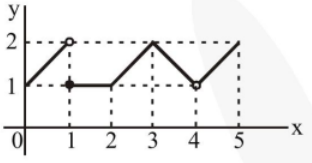
(C) $x f'(x) - f(x) + 1 = 0$

(D) $2 f(x) = f'(x) + 2$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Graph of $f(x)$ is shown in adjacent figure, then in [0, 5]



(A) $f(x)$ has non removable discontinuity at two points

(B) $f(x)$ is non differentiable at four points

(C) $\lim_{x\to 1} f(f(x)) = 1$

(D) Number of points of discontinuity = number of points of non-differentiability

Q2. Which of the following statements is/are correct ?

(A) There exist a function $f : [0,1] \to R$ which is discontinuous at every point in [0,1] & $|f(x)|$ is continuous at every point in [0,1]

(B) Let $F(x) = f(x).\ g(x)$. If $f(x)$ is differentiable at $x = a$, $f(a) = 0$ and $g(x)$ is continuous at $x = a$ then $F(x)$ is always differentiable at $x = a$.

(C) If $f'(a^+) = 2$ & $f'(a^-) = 3$, then $f(x)$ is non-differentiable at $x = a$ but will be always continuous at $x = a$

(D) If $f(a)$ and $f(b)$ possess opposite signs then there must exist at least one solution of the equation $f(x) = 0$ in (a,b) provided f is continuous on [a,b]

Q3. If $f(x) = \text{sgn}(x^5)$, then which of the following is/are **false** (where sgn denotes signum function) -

(A) $f'(0^+) = 1$

(B) $f'(0^-) = -1$

(C) f is continuous but not differentiable at $x = 0$

(D) f is discontinuous at $x = 0$

Q4. Let $f : R \to (0, \pi)$ be defined as

$$f(x) = \cot^{-1}\left(\frac{2-|x|}{2+|x|}\right).$$

Then which of the following statement(s) is (are) true?

(A) f (x) is neither injective nor surjective.

(B) f (x) is continuous and derivable on R.

(C) f (x) is both even function and aperiodic function.

(D) $\lim_{x\to \frac{-2}{\sqrt{3}}} f(x) = \frac{5\pi}{12}$.

Q5. Let $f(x) = \text{Max.}\left(\sin^{-1} x, \frac{3\pi}{4}, \cos^{-1} x\right)$

$\forall\ x \in [-1, 1]$ then -

(A) f(x) is non-derivable at $x = \frac{1}{\sqrt{2}}$.

(B) f(x) is discontinuous at $x = \frac{1}{\sqrt{2}}$.

(C) $f\left(\frac{\pi}{4}\right) = \frac{3\pi}{4}$.

(D) f(x) is continuous but non-derivable at $x = \frac{-1}{\sqrt{2}}$.

Q6. Let S denotes the set of all points where $\sqrt[5]{x^2|x|^3} - \sqrt[3]{x^2|x|} - 1$ is not differentiable then S is a subset of-

(A) {0,1}

(B) {0,1,–1}

(C) {0,1}

(D) {0}

Q7. If function defined by (m > 0)

$$f(x) = \begin{cases} \dfrac{(x-m)}{|x-m|} & , \ x \le 0 \\ 2x^2 + 3ax + b & , \ 0 < x < 1, \\ m^2x + b - 2 & , \ x \ge 1 \end{cases}$$

is continuous & differentiable everywhere, then -

(A) b + m = –1

(B) b + m = 1

(C) b + m = –3

(D) $m^2 + a + b = 3$

Q8. Let $f(x) = (\alpha-1)\sin|x| + (1-\alpha)e^{|x|}$ is differentiable every where, then α can take the value/s is/are-

(A) –7 (B) –3

(C) 4 (D) 6

Q9. Let g(x) = x· f(x), where

$$f(x) = \begin{cases} x\sin\dfrac{1}{x}, & x \ne 0 \\ 0, & x = 0 \end{cases} \text{ at } x = 0.$$

(A) g is differentiable but g' is not continuous

(B) g is differentiable while f is not

(C) Both f and g are differentiable

(D) g is differentiable and g' is continuous

Q10. The function $\phi(x) = [|x| - \sin|x|]$ (where [.] denotes greater integer function) is -

(A) derivable at x = 0

(B) continuous at x = 0

(C) $\lim\limits_{x\to 0}\phi(x)$ does not exists

(D) continuous and derivable at x = 0

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 11 to 13)

Consider $f(x)$ = min (cosx, {x}) in $\left(0, \dfrac{\pi}{2}\right)$ & let x_1, x_2 be the solutions of $\sin^4\dfrac{x}{2} + \{x\} = \cos^4\dfrac{x}{2}$ in $\left(0, \dfrac{\pi}{2}\right)$. Where {x} denotes fractional part function.

If $f'(x_1^-) = \ell$, $f'(x_2^-) = m$ & n is the value of x at which $f(x)$ is discontinuous.

Q11. Value of $[x_1] + [x_2]$ is (where [.] represents greatest integer function) -

(A) 0 (B) 1 (C) 2 (D) 3

Q12. Value of $\ell + m + n$ is equal to -

(A) 0 (B) 1 (C) 2 (D) 3

Q13. Number of points in (0, π/2), where f(x) is not differentiable is equal to -

(A) 1 (B) 2

(C) 3 (D) infinite

Paragraph # 2 (Ques. 11 to 13)

Let $f(x) = f_1(x) - 2\, f_2(x)$

where $f_1(x) = \begin{cases} \min.\left(x^2, |x|\right), & -1 \le x \le 1 \\ x^2, & |x| > 1 \end{cases}$

and $f_2(x) = \begin{cases} |x|, & |x| > 1 \\ \max.\left(x^2, |x|\right), & |x| \le 1 \end{cases}$.

Also $g(x) = \begin{cases} \min.\{f(t) : -3 \le t \le x, \text{ for } -3 \le x < 0\} \\ \max.\{f(t) : 0 \le t \le x, \text{ for } 0 \le x < 3\} \end{cases}$

Q14. Range of $f(x)$, is

(A) $(-\infty, -1]$ (B) $(-\infty, 1]$

(C) $[1, \infty)$ (D) $[-1, \infty)$

Q15. Number of values of x where g (x) fails to be differentiable, is

(A) 0 (B) 1 (C) 2 (D) 3n

Q16. $\lim\limits_{x \to 0} f(g(x))$ is equal to

(A) 0 (B) 1

(C) – 1 (D) non existent.

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17. In the following, [ ] and { } are greatest integer function and fractional part function respectively. Match the functions in column-I with the properties in column-II.

Column-I	**Column-II**
(A) $f_1(x) = \left[\frac{4x}{\pi}\right]$	(P) discontinuous at more than 3 points but less than 6 points in [–2, 2]
(B) $f_2(x) = \cos^{-1}\left(\text{sgn}\left(\cos\frac{2x-1}{2}\pi\right)\right)$	(Q) non derivable at more than 2 points but atmost 5 points in [–2, 2]
(C) $f_3(x) = \max.(\{x+1\}, \{5-x\})$	(R) range contains atleast one integer but not more than seven and no irrational value in [–2, 2]
(D) $f_4(x) = \sqrt{x^2} + [x]^2$	(S) many one but not even function in [–2, 2]
	(T) neither odd nor periodic in [–2, 2]

Q18. **Column-I** **Column-II**

(A) Let f be a real valued differentiable function on R such that $f'(1) = 6$ and $f'(2) = 2$.

then $\lim\limits_{h\to 0} \dfrac{f(3\cosh + 4\sin h - 2) - f(1)}{f(3e^h - 5\sec h + 4) - f(2)}$ is equal to (P) 4

(B) For $a > 0$, let $f : [-4a, 4a] \to R$ be an even function such that $f(x) = f(4a - x)\ \forall\ x \in [2a, 4a]$

and $\lim\limits_{h\to 0} \dfrac{f(2a+h) - f(2a)}{h} = 4$,

then $\lim\limits_{h\to 0} \dfrac{f(h-2a) - f(-2a)}{2h}$ is equal to (Q) 5

(C) Suppose f is a differentiable function on R. Let $F(x) = f(e^x)$ and $G(x) = e^{f(x)}$. If $f'(1) = e^3$ and $f(0) = f'(0) = 3$, then $\dfrac{G'(0)}{F'(0)}$ is equal to (R) 3

(D) Let $f(x) = \text{Max.}(\cos x, x, 2x - 1)$ where $x \geq 0$. Then number of points of non-differentiability of f(x), is equal to (S) 2

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. Let a_1 and a_2 be two values of a for which

$$f(x) = \begin{cases} x.\dfrac{ln(1+x) + ln(1-x)}{\sec x - \cos x}, & x \in (-1,0) \\ (a^2 - 3a + 1)x + x^2, & x \in [0, \infty) \end{cases}$$

is differentiable at $x = 0$, then find the value of $(a_1^2 + a_2^2)$.

Q20. If

$$f(x) = \begin{cases} \dfrac{\sin[|x|]\pi}{a x^4} + a x^2 + b; & -2 \leq x < -1 \\ 3\sec \pi x + \cot^{-1} x; & -1 \leq x \leq \dfrac{-3}{4} \end{cases}$$

is differentiable in $\left(-2, \dfrac{-3}{4}\right)$ then $a = \dfrac{1}{\lambda_1}$ and $b = \dfrac{3\pi - \lambda_2}{4}$. Find the value of $(\lambda_1 + \lambda_2)$.

Note: [y] denotes largest integer $\leq y$.

Q21. If $f(x)$ is a differentiable function such that $f(x + y^2) = f(x) + yf(y)$ and $f'(0) = 2$, $f(0) = 0$, then $(f'(2) + f(3))$ equals -

Q22. Let $f(x)$ be a differentiable function such that $2f(x+y)+f(x-y)=3f(x)+3f(y)+2xy\ \forall\ x, y \in R$ & $f'(0)=0$, then $f(10)+f'(10)$ is equal to

Q23. For any real number x, let [x] denote the largest integer less than or equal to x. Let f be a real valued function defined on the interval $[-3, 3]$ by

$$f(x)=\begin{cases}-x-[-x] & \text{if } [x] \text{ is even}\\ x-[x] & \text{if } [x] \text{ is odd}\end{cases}.$$

If L denotes the number of points of discontinuity and M denotes the number of points of non-derivability of f(x), then find (L + M).

Q24. A derivable function $f: R^+ \to R$ satisfies the condition $f(x)-f(y)\geq \ln(x/y)+x-y$ for every $x, y \in R^+$. If g denotes the derivative of f then compute the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$.

Q25. Let $f(x)=\begin{cases}x^2+1 & ,x<0\\ (x-1)^3+2b & ,x\geq 0\end{cases}$

and $g(x)=\begin{cases}x+2a & ,\text{if } x<0\\ |x-2| & ,\text{if } x\geq 0\end{cases}$

(where $a \in R^-$ and $b \in R$). If fog(x) is continuous for all real x and differentiable at x = 0, then find the value of $8(b-a)$.

Q26. Let f(x) be a differentiable function such that $f'(0)=1$, and the sequence $\{a_n\}$ is defined as $a_1=2$ and $a_n=\lim_{x\to\infty} x^2\left(f\left(\frac{a_{n-1}}{x}\right)-f(0)\right)^2$, $n \in N$. If the value of $\prod_{i=1}^{10} a_i = 2^k$, find k.

SUBJECTIVE TYPE

Q27. Examine the function, $f(x)=x\cdot\frac{a^{1/x}-a^{-1/x}}{a^{1/x}+a^{-1/x}}$, $x\neq 0$ $(a>0)$ and $f(0)=0$ for continuity and existence of the derivative at the origin.

Q28. Let $f(x)=xe^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}$; $x\neq 0$, $f(0)=0$, test the continuity & differentiability at x = 0.

Q29. Discuss the continuity & the derivability in $[0, 2]$ of $f(x)=\begin{cases}|2x-3|[x] & \text{for } x\geq 1\\ \sin\frac{\pi x}{2} & \text{for } x<1\end{cases}$.

Where [.] denote greatest integer function .

Q30. Discuss the continuity and differentiability of $f(x)=\min.\{|x|, |x-2|, 2-|x-1|\}$. Where $\{\cdot\}$ denotes the fractional part function.

Q31. Let $f(0)=0$ and $f'(0)=1$. For a positive integer k, show that -

$$\lim_{x\to 0}\frac{1}{x}\left(f(x)+f\left(\frac{x}{2}\right)+.....f\left(\frac{x}{k}\right)\right)$$

$$=1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{k}$$

Q32. The function

$$f(x)=\begin{cases}ax(x-1)+b & \text{when } x<1\\ x-1 & \text{when } 1\leq x\leq 3\\ px^2+qx+2 & \text{when } x>3\end{cases}.$$

Find the values of the constants a, b, p, q so that

(i) f(x) is continuous for all x

(ii) f'(1) does not exist

(iii) f'(x) is continuous at x = 3

Answer Key

Ex - 1									
1	2	3	4	5	6	7	8	9	10
B	B	A	C	D	D	A	B	D	B
11	12	13							
B	C	C							

Ex - 1A

1. A, C **2.** B, D **3.** B, D **4.** A, B **5.** B, D

6. C 7. D **8.** B **9.** C **10.** A

11. D **12.** (A) $\rightarrow$ (R); (B) $\rightarrow$ (P); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (R)

13. 2 **14.** 3 **15.** 24

16. (a) $f'(1^+) = 1$; $f'(1^-) = -1$ (b) $f'(0^+) = -1$; $f'(0^-) = -1$

17. $a = \frac{-1}{2}$, $b = 1$ **18.** $m \in [0, 1]$

19. f (x) is not differentiable at $x = \frac{1}{2}$, 1 in $x \in (0, 2)$

20. Continous and differentiable at $x = 0$ and $x = \pi/2$

21. f (g (x)) is discontinuous & non derivable at – 1, 0, 1

22. f (x) is not differentiable at $x = \pi - \cos^{-1}\left(\frac{3}{5}\right)$, $\forall\ x \in (0, \pi)$.

23. f(x) is conti. but not derivable at x = 0

24. conti. $\forall$ $x \in R$, not diff. at x = 0,1 & 2

Ex - 2									
1	2	3	4	5	6	7	8	9	10
D	A	A	B	D	B	A	A	A	D
11	12	13	14	15	16				
D	D	B	D	A	A				

Ex - 2A

1. B,C **2.** A,B,C,D **3.** A,B,C **4.** ACD 5. CD

6. A.B,C,D **7.** B,D **8.** A,B,C,D **9.** AB **10.** A,B,D

11. B **12.** D **13.** C **14.** D **15.** C

16. D

17. (A) $\rightarrow$ (P, Q, R, S, T) ; (B) $\rightarrow$ (P, Q, S, T) ; (C) $\rightarrow$ (P, T) ; (D) $\rightarrow$ (P, Q, S, T)

18. (A) $\rightarrow$ (P); (B) $\rightarrow$ (S); (C) $\rightarrow$ (R); (D) $\rightarrow$ (S)

19. 5 **20.** 17 **21.** 8 **22.** 120 **23.** 8

24. 5150 **25.** 15 **26.** 1023

28. f is cont. but not diff. at $x = 0$

29. f is conti. at $x = 1, 3/2$ & disconti. at $x = 2$, f is not diff. at $x = 1, 3/2, 2$.

30. f(x) is continuous $\forall\ x \in R$ and f(x) is not differentiable at $x = \frac{-1}{2}$, 0, 1, 2, $\frac{5}{2}$.

32. $a \neq 1, b = 0, p = \frac{1}{3}$ and $q = -1$.